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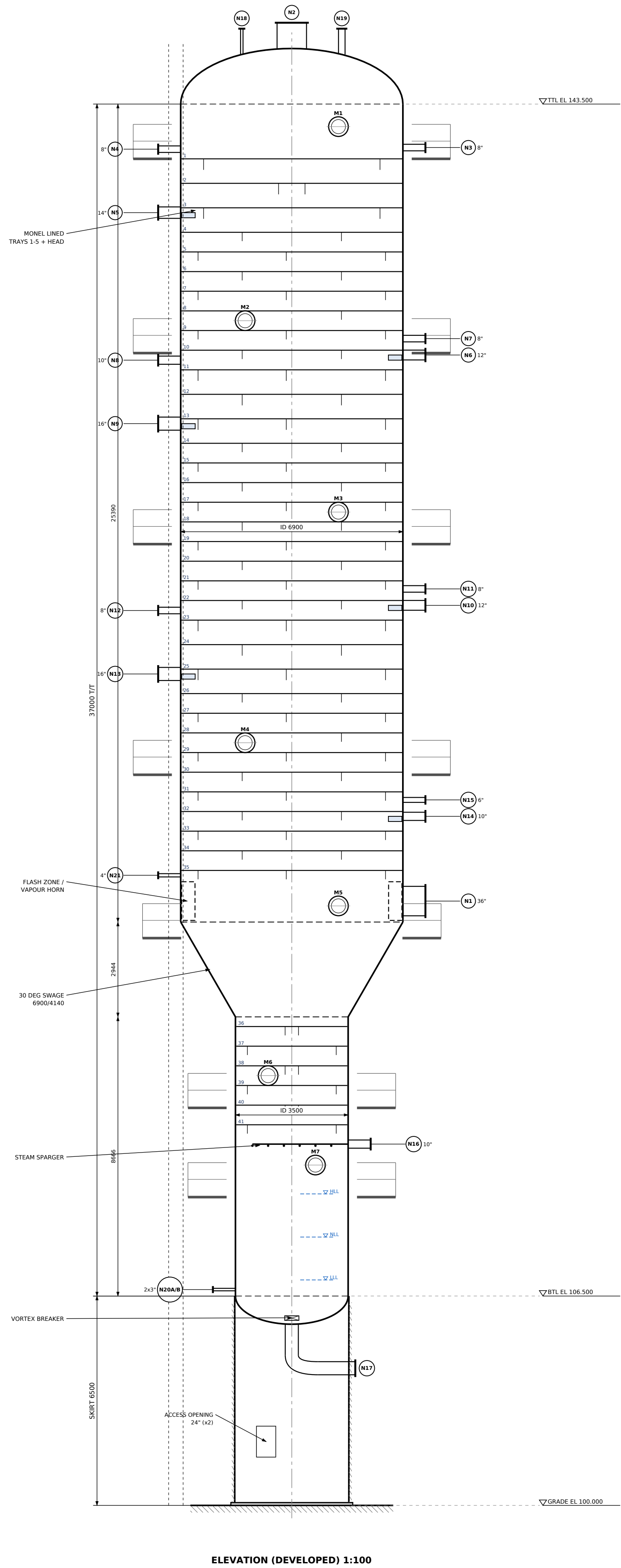
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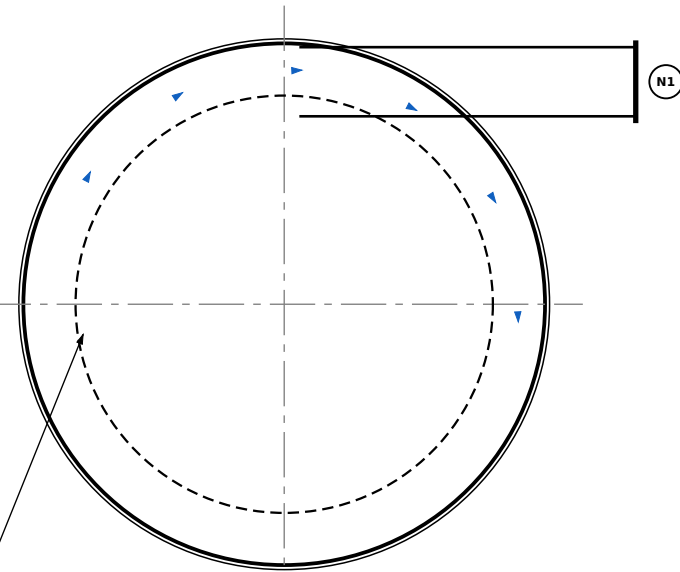
ELEVATION (DEVELOPED) 1:100

NOZZLE SCHEDULE						
MARK	QTY	SIZE	RATING	LOCATION	ORIENT	SERVICE
N1	1	36"	150# RF	EL 118.760	90	Feed - transfer line from H-101 (tangential, vapour horn)
N2	1	36"	150# RF	TOP HEAD	0	Overhead vapour to A-101
N3	1	8"	150# RF	EL 142.150	45	Reflux from P-103
N4	1	8"	150# RF	EL 142.100	270	TPA return
N5	1	14"	150# RF	EL 140.130	225	TPA draw to P-106 (draw sump)
N8	1	10"	150# RF	EL 135.550	270	MPA return
N9	1	16"	150# RF	EL 133.580	225	MPA draw to P-107 (draw sump)
N12	1	8"	150# RF	EL 127.780	270	BPA return
N13	1	16"	150# RF	EL 125.810	225	BPA draw to P-108 (draw sump)
N6	1	12"	150# RF	EL 135.710	135	Kero draw to C-102
N7	1	8"	150# RF	EL 136.220	150	Vapour return from C-102
N10	1	12"	150# RF	EL 127.940	135	Diesel draw to C-103
N11	1	8"	150# RF	EL 128.450	150	Vapour return from C-103
N14	1	10"	150# RF	EL 121.390	135	Agg draw to C-104
N15	1	6"	150# RF	EL 121.900	150	Vapour return from C-104
N16	1	10"	150# RF	EL 111.216	60	Stripping steam (from H-101 SS coil)
N17	1	16"	150# RF	BTM HEAD	0	Atm. residue to P-112
N18	1	3"	150# RF	TOP HEAD	0	Vent / steam-out
N19	4	8"	150# RF	TOP HEAD	0	PSV-1001 relief (4 x T orifice)
N20A/B	2	3"	150# RF	EL 106.700	300	LT / LG bridle (bottom sump)
N21	1	4"	150# RF	EL 119.560	200	Wash-zone overflow / slop connection (spare)
M1	1	24"	150# RF	EL 142.800	0	Manway 24" (davitt)
M2	1	24"	150# RF	EL 136.775	0	Manway 24" (davitt)
M3	1	24"	150# RF	EL 130.835	0	Manway 24" (davitt)
M4	1	24"	150# RF	EL 123.675	0	Manway 24" (davitt)
M5	1	24"	150# RF	EL 118.610	0	Manway 24" (davitt)
M6	1	24"	150# RF	EL 113.341	0	Manway 24" (davitt)
M7	1	24"	150# RF	EL 110.566	0	Manway 24" (davitt)

TRAY HYDRAULIC SECTIONS (PROCESS SIZING; % FLOOD AT SELECTED ID, 85 % NET AREA)										
SECTION	TRAYS	TS mm	V 4th	L 1th	FLV	u flooded	D calc	D sel.	PASSES	% FLOOD
Top / TPA	1-5	760	229	268	0.106	1.12	4.60	6.90	2	36
Naphtha-kero fract.	4-9	610	335	207	0.057	1.02	5.51	6.90	4	51
MPA	11-13	760	620	438	0.079	0.94	6.74	6.90	4	76
Kero-diesel fract.	14-21	610	564	339	0.069	0.81	6.68	6.90	4	75
BPA	23-25	760	710	384	0.068	0.85	6.85	6.90	4	79
Diesel-AGO fract.	26-31	610	529	275	0.065	0.75	6.23	6.90	4	65
Wash zone	33-35	610	339	34	0.010	1.07	5.24	6.90	4	46
Stripping	36-41	610	15	257	0.796	0.82	2.47	3.50	2	40

SHELL COURSE / THICKNESS SCHEDULE (CFU-000-ME-CAL-001)						
COMPONENT	ID mm	FROM EL	TO EL	t mm	MATERIAL	REMARK
Skirt	3544 OD	100.000	106.500	20	SA-516-70	fireproofed
Bottom head 2:1	3500	106.500	106.500	12	SA-516-70	
Stripping section	3500	106.500	115.166	22	SA-516-70	t req 9.0
Swage 30 deg	6900 cone	115.166	118.110	20	SA-516-70	t req 15.8
Flash zone / rectifying se	6900	118.110	143.500	16	SA-516-70	t req 14.1
Top head 2:1	6900	143.500	-	16	SA-516-70	

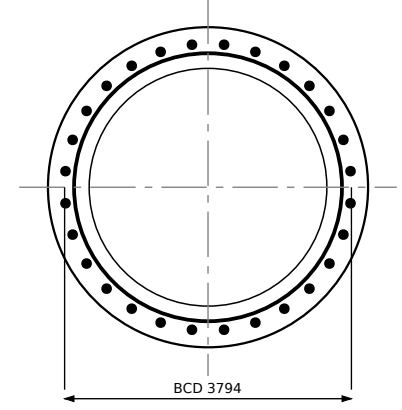
PLATFORM SCHEDULE (1200 WIDE, 60 % WRAP)			
PLATF.	ELEVATION	SERVES MANWAY	NOZZLES ACCESSED
P1	EL 109.570	M7	N16
P2	EL 112.340	M6	
P3	EL 117.610	M5	N1, N21
P4	EL 122.680	M4	
P5	EL 129.840	M3	
P6	EL 135.770	M2	N8, N6, N7
P7	EL 141.800	M1	N3, N4
P8	EL 145.425	top head / davitt	



SECTION THROUGH FLASH ZONE 1:100

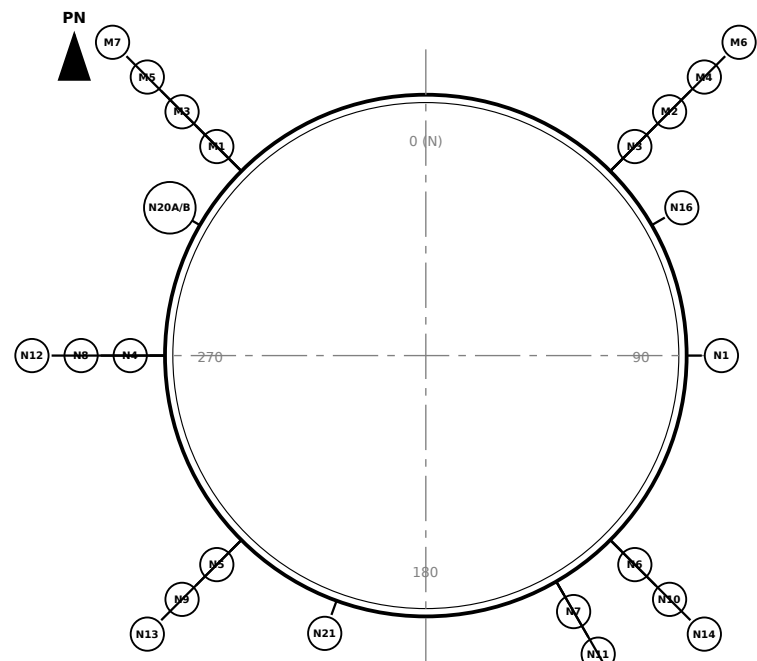
tangential feed, 270 deg horn

SKIRT BASE / ANCHOR BOLT PLAN 1:100



28 x M64 ASTM F1554 Gr.55

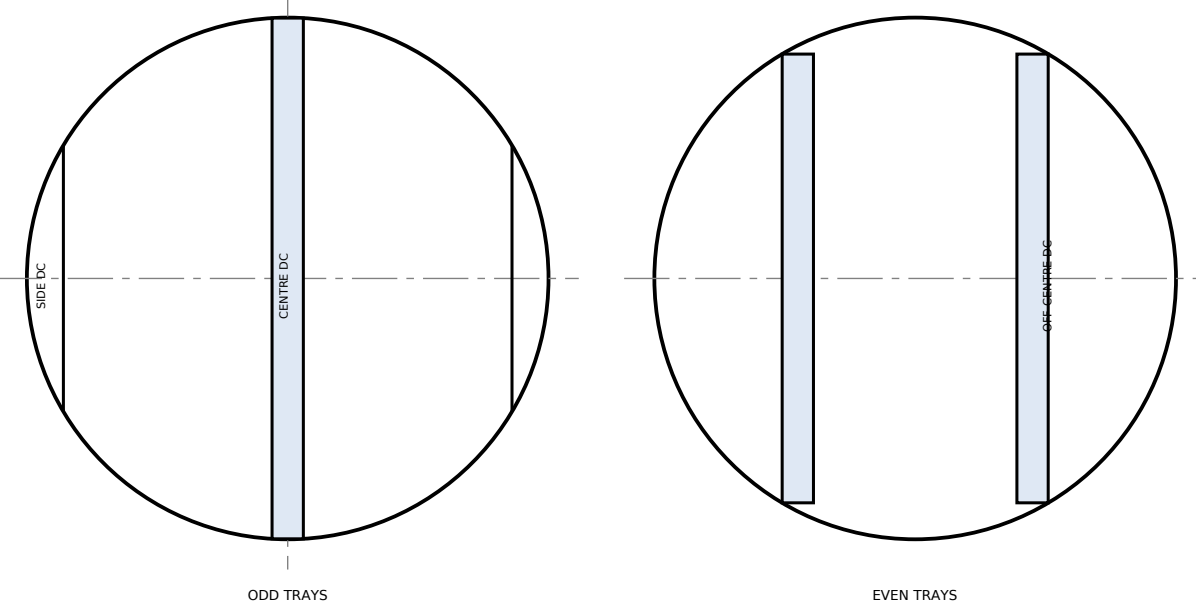
skirt OD 3544 x 20 mm; base ring by vendor



NOZZLE ORIENTATION PLAN 1:100

(manways alternate 315 / 45 deg - platform side)

TYPICAL 4-PASS TRAY ARRANGEMENT (ID 6900) 1:100



TRAY SPACING (mm)	
Trays 1-3, 11-13, 23-25 (PA)	760
Other rectifying trays	610
Flash zone (tray 35-36)	4844
Stripping trays 36-41	610
Top TL to tray 1	1700

DESIGN DATA	
Service	Atmospheric crude fractionator
Design pressure	3.5 barg
Design temperature	390 C
Operating P / T	1.19 / 1.47 / 134 / 361 C
Shell material	SA-516-70
Cladding / lining	C5 + 4105 clad below tray 10; Monel 400 lined top head & tra
Corrosion allowance	3.0 mm
Joint efficiency	1.0 (full RT)
Shell thickness	22, 20, 16 mm (by course, bottom up)
Heads (2:1 SD) top / btm	16 / 12 mm
Skirt OD x t x height	3544 x 20 x 6500 mm
Anchor bolts	28 x M64 on 3794 BCD
Hydrotest (top / btm)	5.74 / 9.63 barg
Insulation	125 mm mineral wool
Weight empty / oper.	330 / 405 t
Weight hydrotest	1458 t
Wind base V / M	801 kN / 21781 kNm
Code	ASME VIII Div.1, U-stamp; ASCE 7-16

LEGEND	
	Valve tray with downcomer
	Liquid distributor
	Liquid level (LLL/NLL/HLL)
	Nozzle - mark and size
	Draw sump
	Packed bed (structured / grid)
	Collector / chimney tray
	Manway (shown on face)
	Platform with handrail
	External stiffening ring

WEIGHTS & LOADS	
Shell + heads + nozzles + clad	127.1 t
Skirt + base ring	14.4 t
Internals	116.9 t
Platforms / ladders	37.7 t
Insulation + fireproofing	33.4 t
EMPTY (installed)	329.6 t
OPERATING	405.0 t
HYDROTEST (full of water)	1458.2 t
Wind base shear / moment (ASCE 7, strength)	801 kN / 21781 kNm
Natural frequency / gust factor	0.77 Hz / 1.16

- NOTES**
- Dimensions in mm, elevations in m. Grade = EL 100.000.
 - Thicknesses are FEED minimum nominal (CFU-000-ME-CAL-001); fabricator to verify.
 - Nozzle orientations preliminary - to be confirmed by piping study.
 - Platforms 1200 wide, 60 % wrap; caged ladders not shown in full.
 - Datasheet CFU-000-ME-DS-001; weights in data/mech.json.
 - Trays 1-5 and top head Monel 400 lined; 4105 clad below tray 10.
 - Swage 30 deg half-angle in flash zone; tray 36 300 below junction.
 - Draw sumps (shaded) at trays 3, 10, 13, 22, 25, 32.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
COLUMN GENERAL ARRANGEMENT C-101 ATMOSPHERIC CRUDE FRACTIONATOR			
DISCIPLINE	STAGE	SCALE	SHEET
MECHANICAL	FEED	1:100	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-ME-GA-001			A